

Flow Matching versus Denoising Diffusion on 2D Toy Distributions: Few-Step Sample Quality and What One Reflow Round Adds

Altair-98J

Stemma seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/flow-matching-vs-diffusion-2d>

Abstract—We compare denoising diffusion (DDPM with DDIM and ancestral samplers), conditional flow matching with Euler and Heun samplers, and one-round rectified-flow reflow on three 2D toy densities, using one shared MLP, one training budget, five seeds, and sliced Wasserstein and MMD scores against 10000 held-out points for 1 to 100 sampling steps. All methods are reimplemented. Flow matching has lower one-step sliced Wasserstein (SW) than DDIM on every task but higher one-step MMD, and neither is usable at one step. At four steps it has lower SW on eight Gaussians, marginally lower SW on the checkerboard, and higher SW on two moons; this reversal is SW-only, since MMD ranks flow matching ahead of DDIM at four steps on all three tasks. Whether its advantage shrinks with more steps is mixed: the SW ratio to DDIM shrinks on eight Gaussians, grows on the checkerboard and changes side on two moons. Flow-matching trajectories are measured as less straight than DDIM trajectories by chord-to-arc length. One reflow round brings one-step SW close to the real-data floor (0.042 against 0.022 on eight Gaussians, where the gap is widest) and gives no benefit at 100 steps; the student inherits the quality of its teacher’s sampler (a 4-step teacher limits it at every reported step count), and a second round adds little. Without retraining, a Heun sampler on the flow-matching model is better than Euler at 8 network evaluations on two moons and the checkerboard and worse at 2. In an ablation, cosine-schedule DDIM fails at every tested step count unless its sampling grid is started below the last timestep. The study is small (2D, one network size, 3000 updates, no tuning), and the reflow comparison is not matched in training compute.

Index Terms—flow matching, denoising diffusion, rectified flow, reflow, few-step sampling, sliced Wasserstein distance

I. Introduction

Denoising diffusion models [1], [2] and flow matching [3]–[5] both turn Gaussian noise into data by integrating a learned vector field, yet they are trained on different paths. Diffusion models regress the noise added along a variance-preserving schedule, and the resulting probability-flow trajectories are typically curved, so deterministic samplers such as DDIM [6] need tens of steps. Flow matching regresses the velocity of a straight line between a noise sample and a data sample. The

common argument is that straighter marginal paths can be integrated accurately with few Euler steps, and that rectified flow [4] can straighten them further by reflow: retraining on (noise, sample) couplings produced by the model itself.

The published evidence for this argument is mostly on image benchmarks. Lipman et al. [3] compare flow matching with diffusion-path and score-matching baselines on CIFAR-10 and ImageNet using one shared architecture, and report likelihood, FID and the number of function evaluations of an adaptive solver; Liu et al. [4] report FID and recall of one-step rectified flows on CIFAR-10. Both papers, and minibatch-OT flow matching [7], also show 2D toy examples. We do not question that evidence. We run a deliberately small, controlled replication on three 2D densities (eight Gaussians, two moons, checkerboard), a setting in which fresh samples of the target are available, so that distributional distances to held-out data can be computed directly, every step count from 1 to 100 can be swept, a DDIM sampler can serve as the diffusion baseline, and seed-to-seed variance can be reported. Every system uses the same MLP, optimiser, number of updates and training data stream, and is scored with the same held-out reference, a sliced Wasserstein distance [8], [9] and an MMD, at 1 to 100 sampling steps. We ask: (H1) does flow matching give better samples than DDIM at four steps; (H2) does that advantage vanish with more steps; (H3) does one reflow round help one-step sampling without helping at 100 steps; (H4) are flow trajectories actually straighter, as measured by chord-to-arc length.

Contributions: a single-file CPU reimplementations of DDPM/DDIM, flow matching and reflow on a shared network; five-seed step-count curves with a real-data floor; an equal-evaluation comparison of Heun and Euler sampling; ablations on the diffusion schedule, the reflow teacher and a second round; and a hypothesis-by-hypothesis report including failures. The study is small (2D, one MLP, 3000 updates) and its conclusions are limited accordingly (Sec. VII).

Altair-98J is the codename of an autonomous study, not a person. Produced by the Stemma autonomous AI research pipeline; the topic was chosen to cover the field. Every reported number comes from logged runs in the linked repository and every reference was checked against arXiv, Crossref, DataCite or OpenAlex.

II. Related Work

Diffusion models and fast samplers. DDPM [1] trains a network to predict the noise added by a fixed forward process; Song et al. [2] unify this with score-based models through SDEs and a probability-flow ODE. DDIM [6] observes that the same training objective supports a family of non-Markovian samplers with a tunable noise level η on a sub-sampled timestep grid, which is the few-step diffusion baseline we use ($\eta=0$ and $\eta=1$). Improved DDPM [10] introduced the cosine schedule that we use in an ablation, and EDM [11] studies how preconditioning, noise levels and solvers interact in sampling cost; we do not reimplement EDM.

Flow matching and rectified flow. Flow matching [3] regresses a conditional vector field and obtains simulation-free training of continuous normalising flows. Rectified flow [4] proposes the straight interpolation and the reflow procedure, and stochastic interpolants [5] give a unifying view; the flow matching guide [12] collects the formulations and practice. Minibatch optimal-transport couplings [7] straighten paths at training time instead of through reflow. Theory of reflow is developed in [13], and Rectified Diffusion [14] argues that what matters for few-step quality is that the training target is a first-order ODE with a pre-trained pairing, not straightness itself; our straightness measurements and the diffusion-schedule ablation bear on this question only in 2D. Consistency models [15] distil a pretrained model, or are trained directly, for one-step generation, which we do not study.

Metrics. Sliced Wasserstein distances [8], [9] average one-dimensional Wasserstein distances over random projections and are cheap in 2D; we use the sliced W_1 with a fixed set of directions together with a kernel MMD, and report the distance between fresh true samples and the reference as a noise floor.

III. Method

All systems share one velocity/noise network $f_\theta(x, t)$: an MLP with four hidden layers of width 128, SiLU activations, and as input $x \in \mathbb{R}^2$, the scalar $t \in [0, 1]$ and the sin/cos Fourier features of t at eight frequencies $2^j\pi$, $j = 0, \dots, 7$. Let p be the data density and $z \sim \mathcal{N}(0, I)$.

DDPM (reimplemented). With $T=1000$ steps, β_i linear from 10^{-4} to 0.02 and $\bar{\alpha}_i = \prod_{j \leq i} (1 - \beta_j)$, we draw $x_0 \sim p$, $i \sim \mathcal{U}\{0, \dots, T-1\}$, $\epsilon \sim \mathcal{N}(0, I)$, set $x_i = \sqrt{\bar{\alpha}_i}x_0 + \sqrt{1 - \bar{\alpha}_i}\epsilon$ and minimise

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E} \|f_\theta(x_i, i/T) - \epsilon\|^2. \quad (1)$$

DDIM sampler (reimplemented). On the uniform sub-grid $\tau_k = \lfloor (T-1)(1 - (k-1)/K) \rfloor$, $k = 1, \dots, K$, of timestep indices (the last step goes to $\bar{\alpha}=1$), $\hat{x}_0 = (x - \sqrt{1 - \bar{\alpha}_\tau}\epsilon)/\sqrt{\bar{\alpha}_\tau}$ and

$$x' = \sqrt{\bar{\alpha}_{\tau'}}\hat{x}_0 + \sqrt{1 - \bar{\alpha}_{\tau'} - \sigma^2}\hat{\epsilon} + \sigma\xi, \quad \sigma = \eta\sqrt{\frac{1 - \bar{\alpha}_{\tau'}}{1 - \bar{\alpha}_\tau}}\sqrt{1 - \frac{\bar{\alpha}_\tau}{\bar{\alpha}_{\tau'}}} \quad (2)$$

with $\hat{\epsilon} = f_\theta(x, \tau/T)$ and $\xi \sim \mathcal{N}(0, I)$. $\eta=0$ is “DDIM” and $\eta=1$ is the strided ancestral DDPM sampler (“DDPM ancestral”). No clipping of $\hat{x}_0$ is applied. In the cosine ablation $\bar{\alpha}$ follows [10] (offset 0.008, $\beta \leq 0.999$), and the diagnostic variant replaces $T-1$ in the grid by the last index with $\bar{\alpha} \geq 4 \times 10^{-5}$.

Conditional flow matching (reimplemented). With noise at $t=0$ and data at $t=1$, $x_t = (1-t)z + tx_1$, $x_1 \sim p$, $t \sim \mathcal{U}[0, 1]$:

$$\mathcal{L}_{\text{FM}} = \mathbb{E} \|f_\theta(x_t, t) - (x_1 - z)\|^2. \quad (3)$$

Sampling integrates $\dot{x} = f_\theta(x, t)$ from $t=0$ to 1 with K uniform steps of Euler, $x \leftarrow x + \Delta f_\theta(x, t)$, $\Delta = 1/K$, or Heun, $x \leftarrow x + \frac{\Delta}{2}(f_\theta(x, t) + f_\theta(\tilde{x}, t + \Delta))$, $\tilde{x} = x + \Delta f_\theta(x, t)$, which costs two network evaluations per step.

Reflow (reimplemented). Given a trained flow model $f^{(r-1)}$, draw $N=20000$ noise samples z_n and compute $x_n = \text{ODE}_{100}(z_n; f^{(r-1)})$ with 100 Euler steps (4 in one ablation). The student $f^{(r)}$ is initialised from $f^{(r-1)}$ and trained with $\mathcal{L}_{\text{FM}}$ on the fixed pairs (z_n, x_n) , $x_t = (1-t)z_n + tx_n$, with the same optimiser and number of updates. “Reflow-1” uses the FM model as teacher; “Reflow-2” uses Reflow-1.

Metrics. For $n=10000$ generated points x_i and the 10000 held-out points y_i , the sliced W_1 is $\text{SW} = \frac{1}{L} \sum_{l=1}^L \frac{1}{n} \sum_i |x_{(i)}^\top u_l - y_{(i)}^\top u_l|$ with $L=256$ fixed unit directions u_l and sorted projections. MMD is the biased estimate $\text{MMD}^2 = k(x, x') + k(y, y') - 2k(x, y)$ with k the mean of four RBF kernels (bandwidths 0.1, 0.3, 1, 3) on the first 3000 points of each set. Straightness is the mean over 2000 samples of $\|x_{\text{end}} - x_{\text{start}}\| / \sum_k \|x_{k+1} - x_k\|$ along the 100-step deterministic path of the system’s own sampler (Euler or Heun for flows, DDIM for diffusion); it equals 1 for a straight line. It is a scale-free chord-to-arc ratio and not the straightness functional $\int_0^1 \mathbb{E} \|(Z_1 - Z_0) - \dot{Z}_t\|^2 dt$ of [4].

IV. Experimental Setup

Tasks and data. Eight Gaussians: eight components with std 0.1 on a circle of radius 2. Two moons: two half circles with Gaussian noise 0.05, centred and scaled by 2. Checkerboard: uniform on the black cells of a 4×4 board on $[-2, 2]^2$. Training batches are sampled on-line; the held-out reference is 10000 points from a fixed generator seed that is never used for training.

Training. All networks: same MLP, Adam with learning rate 2×10^{-3} cosine-annealed to 0, 3000 updates, batch 512, EMA of the weights (decay 0.999 with warm-up), and the EMA weights are used for sampling. Reflow students use the same settings on a pool of 20000 pairs. These values were chosen once, before any run, and not tuned for any system; the reduction from 5000 to 3000 updates was made before the main runs to fit the CPU budget (recorded with rh decide). The tuning budget of every system is therefore zero.

Evaluation. Step counts $K \in \{1, 2, 4, 8, 16, 32, 64, 100\}$ on uniform grids, the same for every system; one network evaluation per step for Euler, DDIM and ancestral sampling and two for Heun. Seeds 0–4 (main) and 0–2 (ablations); the seed fixes initialisation, training stream and sampling noise, and all systems of a seed share the sampling noise generator seed. Paired tests use the seed as pairing key; all reported tests are two-sided paired t -tests without multiplicity correction, computed from the run registry by the harness’s `rh compare`, which compares two systems of one run group on one metric. Every result number in the text is printed from the run registry at build time. “Real data” is a fresh set of 10000 true samples scored against the reference with the same metrics and gives the floor reachable with a perfect sampler. All runs are on a shared laptop CPU with two threads; one run (training plus the 8 step counts) takes about 3 to 54 s, and the registered runs, launched two at a time, were completed within about 35 minutes of wall-clock. Code and logs: `method/run.py`, `results/runs.jsonl`.

V. Results

Tables I and II give means and standard deviations over five seeds; Fig. 1 plots SW against steps; Table III lists the registered tests and further paired tests that were not registered (two-sided paired t -tests over seeds from `rh compare`, $\alpha = 0.05$, no multiplicity correction, five pairs, so the p values are indicative only). The “Real data” rows are the metric floor: no sampler can be expected below them.

H1: flow matching beats DDIM at four steps — partly supported; reversed on two moons. At $K=4$ FM-Euler has lower SW than DDIM on eight Gaussians (Table I; $p = 0.0005$ in Table III) and on the checkerboard ($p = 0.007$), but the checkerboard gap (0.013) is about one DDIM seed standard deviation, and on two moons DDIM has the lower SW (0.203 against 0.263 for FM, $p = 0.002$; the same direction nominally at $K=2, 8$ and 16, where the $K=16$ gap is small). This reversal is specific to SW: the MMD at $K=4$ on the same runs favours FM on every task, including two moons (0.0145 against 0.0252, Table II; paired p of 0.001, 8.2×10^{-5} and 0.0003 on the three tasks), so by MMD H1 holds on all three tasks and the two-moons failure is metric-dependent. The registered refutation rule (DDIM not worse on at least two tasks) is not triggered, but “flow matching gives better few-step samples” is not a general statement in this study. At $K=1$ FM has the lower SW on all tasks, yet its MMD is higher than that of DDIM on all three, so the two metrics disagree there; we have not isolated why. DDPM ancestral and DDIM coincide at $K=1$ (the last step has no noise). At $K=4$ ancestral sampling is worse than DDIM in SW and MMD on two moons and the checkerboard (paired p at most 1.5×10^{-5} , Table III); on eight Gaussians both differences are within seed noise (SW 0.329 ± 0.015 against $0.288 \pm$

0.043; the MMD difference DDIM–ancestral of -0.0087 is about one DDIM seed standard deviation in size).

H2: the gap closes with more steps — nominally refuted; mixed. The SW ratio DDIM/FM (Table III) moves towards 1 between $K=4$ and $K=100$ only on eight Gaussians; on the checkerboard it moves away from 1, and on two moons it changes from DDIM-better to FM-better, with $|r-1|$ (the rule as registered) nominally larger at $K=100$ (0.77 to 1.25), although a log-ratio would call it slightly closer. At $K=100$ FM has the lower mean SW on all tasks, but the evidence differs by task. On eight Gaussians DDIM’s seed-to-seed spread is about five times that of FM (0.057 ± 0.028 against 0.027 ± 0.005) and the gap is not significant ($p = 0.086$); on two moons the spreads are similar and the FM–DDIM difference of -0.006 is marginal ($p = 0.048$); on the checkerboard the gap is consistent across seeds (0.053 ± 0.002 against 0.031 ± 0.003 , $p = 9.7 \times 10^{-5}$), so there DDIM has not reached the FM error by 100 steps.

H3: one reflow round — supported. Reflow-1 lowers one-step SW on every task by a large margin (for example 0.042 against 1.234 on eight Gaussians, $p = 9.7 \times 10^{-10}$) and stays close to the real-data floor at every K of Table I on every task, while its MMD stays above the floor (0.0016 against 0.0005 on the checkerboard at $K=4$). At $K=100$ it does not improve: the FM–Reflow difference of SW is -0.0010 , -0.0012 and -0.0003 on the three tasks, that is, Reflow-1 is slightly worse, with $p = 0.020$, 0.008 and 0.0002. Reflow-1 received twice the updates of FM-Euler plus 20000 teacher trajectories (no FM control).

H4: straightness — first part refuted, second part supported. By chord/arc length FM paths (0.6565, 0.6763, 0.4500 for the three tasks) are less straight than DDIM paths (0.7524, 0.8142, 0.7770; paired p at most 6.7×10^{-5}), while Reflow-1 paths are at 0.9990, 0.9995 and 0.9988. FM is less straight than DDIM on all three tasks, yet has lower SW at $K=4$ on two of them and lower MMD on all three, so straightness by this measure does not order the two families by few-step quality. Among the Euler-sampled flow models of Table I, the straighter one (Reflow-1) is the better model at $K \leq 4$; the 4-step-teacher student of Sec. VI shows that the converse does not hold. The measure compares trajectories of different parametrisations (a DDIM path in x -space includes its $\sqrt{\alpha}$ rescaling), so this refutation is specific to the measure we registered.

Heun at equal evaluations (exploratory). FM-Heun has the lowest SW in Table I among the systems without a second training stage at every K from 1 to 16, but it uses $2K$ network evaluations, so it must be compared with the $2K$ columns of the other samplers. These comparisons are between seed means of Table I at different step counts; no paired test is reported for them. At 8 evaluations Heun ($K=4$) has lower SW than FM-Euler ($K=8$) on two moons (0.041 against 0.125) and the checkerboard (0.039 against 0.113), and by a small margin on eight Gaussians (0.046

TABLE I

Sliced W_1 against held-out samples (mean $\pm$ std over 5 seeds, lower is better) for K sampling steps; n is the number of seeds. Heun uses $2K$ network evaluations, all others K . The “Real data” rows score 10000 fresh true samples against the same reference.

Method	Task	SW@1 $\downarrow$	SW@2 $\downarrow$	SW@4 $\downarrow$	SW@8 $\downarrow$	SW@16 $\downarrow$	SW@100 $\downarrow$	n
DDIM	8 Gaussians	5.101 $\pm$ 1.386	0.652 $\pm$ 0.049	0.288 $\pm$ 0.043	0.106 $\pm$ 0.052	0.069 $\pm$ 0.030	0.057 $\pm$ 0.028	5
DDPM ancestral	8 Gaussians	5.101 $\pm$ 1.386	0.649 $\pm$ 0.055	0.329 $\pm$ 0.015	0.104 $\pm$ 0.016	0.069 $\pm$ 0.020	0.044 $\pm$ 0.018	5
Flow Matching (Euler)	8 Gaussians	1.234 $\pm$ 0.004	0.417 $\pm$ 0.008	0.086 $\pm$ 0.007	0.054 $\pm$ 0.007	0.044 $\pm$ 0.013	0.027 $\pm$ 0.005	5
Flow Matching (Heun)	8 Gaussians	0.719 $\pm$ 0.128	0.138 $\pm$ 0.011	0.046 $\pm$ 0.005	0.037 $\pm$ 0.007	0.038 $\pm$ 0.013	0.026 $\pm$ 0.005	5
Reflow-1 (Euler)	8 Gaussians	0.042 $\pm$ 0.006	0.033 $\pm$ 0.003	0.035 $\pm$ 0.006	0.035 $\pm$ 0.005	0.036 $\pm$ 0.011	0.028 $\pm$ 0.005	5
Real data (floor)	8 Gaussians	0.022 $\pm$ 0.008	0.022 $\pm$ 0.008	0.022 $\pm$ 0.008	0.022 $\pm$ 0.008	0.022 $\pm$ 0.008	0.022 $\pm$ 0.008	5
DDIM	Two moons	4.248 $\pm$ 0.395	0.438 $\pm$ 0.044	0.203 $\pm$ 0.018	0.101 $\pm$ 0.011	0.056 $\pm$ 0.006	0.031 $\pm$ 0.007	5
DDPM ancestral	Two moons	4.248 $\pm$ 0.395	0.457 $\pm$ 0.073	0.455 $\pm$ 0.014	0.232 $\pm$ 0.009	0.108 $\pm$ 0.006	0.030 $\pm$ 0.006	5
Flow Matching (Euler)	Two moons	1.145 $\pm$ 0.008	0.591 $\pm$ 0.011	0.263 $\pm$ 0.012	0.125 $\pm$ 0.006	0.065 $\pm$ 0.006	0.024 $\pm$ 0.006	5
Flow Matching (Heun)	Two moons	0.763 $\pm$ 0.051	0.149 $\pm$ 0.007	0.041 $\pm$ 0.012	0.028 $\pm$ 0.007	0.031 $\pm$ 0.010	0.026 $\pm$ 0.008	5
Reflow-1 (Euler)	Two moons	0.028 $\pm$ 0.002	0.028 $\pm$ 0.003	0.029 $\pm$ 0.006	0.031 $\pm$ 0.009	0.032 $\pm$ 0.010	0.026 $\pm$ 0.006	5
Real data (floor)	Two moons	0.022 $\pm$ 0.007	0.022 $\pm$ 0.007	0.022 $\pm$ 0.007	0.022 $\pm$ 0.007	0.022 $\pm$ 0.007	0.022 $\pm$ 0.007	5
DDIM	Checkerboard	2.408 $\pm$ 0.241	0.501 $\pm$ 0.039	0.232 $\pm$ 0.010	0.130 $\pm$ 0.008	0.082 $\pm$ 0.005	0.053 $\pm$ 0.002	5
DDPM ancestral	Checkerboard	2.408 $\pm$ 0.241	0.571 $\pm$ 0.027	0.355 $\pm$ 0.004	0.208 $\pm$ 0.002	0.118 $\pm$ 0.002	0.058 $\pm$ 0.004	5
Flow Matching (Euler)	Checkerboard	0.931 $\pm$ 0.005	0.417 $\pm$ 0.007	0.219 $\pm$ 0.005	0.113 $\pm$ 0.010	0.064 $\pm$ 0.006	0.031 $\pm$ 0.003	5
Flow Matching (Heun)	Checkerboard	0.541 $\pm$ 0.008	0.079 $\pm$ 0.005	0.039 $\pm$ 0.010	0.037 $\pm$ 0.011	0.035 $\pm$ 0.004	0.030 $\pm$ 0.003	5
Reflow-1 (Euler)	Checkerboard	0.032 $\pm$ 0.003	0.030 $\pm$ 0.002	0.034 $\pm$ 0.007	0.037 $\pm$ 0.009	0.034 $\pm$ 0.005	0.031 $\pm$ 0.003	5
Real data (floor)	Checkerboard	0.014 $\pm$ 0.000	0.014 $\pm$ 0.000	0.014 $\pm$ 0.000	0.014 $\pm$ 0.000	0.014 $\pm$ 0.000	0.014 $\pm$ 0.000	5

TABLE II

MMD² (lower is better) at four step counts and the straightness (chord/arc length of 100-step paths, higher is straighter; not defined for the stochastic sampler or for real data), same runs as Table I.

Method	Task	MMD@1 $\downarrow$	MMD@4 $\downarrow$	MMD@16 $\downarrow$	MMD@100 $\downarrow$	straight $\uparrow$	n
DDIM	8 Gaussians	0.1937 $\pm$ 0.0284	0.0394 $\pm$ 0.0089	0.0055 $\pm$ 0.0040	0.0032 $\pm$ 0.0032	0.7524 $\pm$ 0.0128	5
DDPM ancestral	8 Gaussians	0.1937 $\pm$ 0.0284	0.0480 $\pm$ 0.0034	0.0049 $\pm$ 0.0033	0.0025 $\pm$ 0.0027	–	5
Flow Matching (Euler)	8 Gaussians	0.6835 $\pm$ 0.0123	0.0053 $\pm$ 0.0004	0.0010 $\pm$ 0.0002	0.0005 $\pm$ 0.0002	0.6565 $\pm$ 0.0016	5
Flow Matching (Heun)	8 Gaussians	0.1351 $\pm$ 0.0250	0.0016 $\pm$ 0.0002	0.0007 $\pm$ 0.0002	0.0005 $\pm$ 0.0002	0.6616 $\pm$ 0.0017	5
Reflow-1 (Euler)	8 Gaussians	0.0024 $\pm$ 0.0002	0.0010 $\pm$ 0.0002	0.0008 $\pm$ 0.0002	0.0006 $\pm$ 0.0002	0.9990 $\pm$ 0.0004	5
Real data (floor)	8 Gaussians	0.0004 $\pm$ 0.0001	0.0004 $\pm$ 0.0001	0.0004 $\pm$ 0.0001	0.0004 $\pm$ 0.0001	–	5
DDIM	Two moons	0.1806 $\pm$ 0.0126	0.0252 $\pm$ 0.0012	0.0020 $\pm$ 0.0003	0.0009 $\pm$ 0.0003	0.8142 $\pm$ 0.0034	5
DDPM ancestral	Two moons	0.1806 $\pm$ 0.0126	0.0486 $\pm$ 0.0024	0.0030 $\pm$ 0.0005	0.0007 $\pm$ 0.0001	–	5
Flow Matching (Euler)	Two moons	0.5620 $\pm$ 0.0182	0.0145 $\pm$ 0.0009	0.0015 $\pm$ 0.0003	0.0008 $\pm$ 0.0001	0.6763 $\pm$ 0.0037	5
Flow Matching (Heun)	Two moons	0.1313 $\pm$ 0.0161	0.0012 $\pm$ 0.0003	0.0008 $\pm$ 0.0002	0.0008 $\pm$ 0.0001	0.6832 $\pm$ 0.0035	5
Reflow-1 (Euler)	Two moons	0.0010 $\pm$ 0.0001	0.0008 $\pm$ 0.0002	0.0008 $\pm$ 0.0002	0.0008 $\pm$ 0.0001	0.9995 $\pm$ 0.0001	5
Real data (floor)	Two moons	0.0005 $\pm$ 0.0001	0.0005 $\pm$ 0.0001	0.0005 $\pm$ 0.0001	0.0005 $\pm$ 0.0001	–	5
DDIM	Checkerboard	0.1368 $\pm$ 0.0211	0.0206 $\pm$ 0.0010	0.0042 $\pm$ 0.0003	0.0033 $\pm$ 0.0002	0.7770 $\pm$ 0.0102	5
DDPM ancestral	Checkerboard	0.1368 $\pm$ 0.0211	0.0383 $\pm$ 0.0011	0.0061 $\pm$ 0.0003	0.0036 $\pm$ 0.0003	–	5
Flow Matching (Euler)	Checkerboard	0.5422 $\pm$ 0.0157	0.0139 $\pm$ 0.0005	0.0025 $\pm$ 0.0004	0.0015 $\pm$ 0.0002	0.4500 $\pm$ 0.0055	5
Flow Matching (Heun)	Checkerboard	0.0915 $\pm$ 0.0034	0.0017 $\pm$ 0.0003	0.0016 $\pm$ 0.0003	0.0015 $\pm$ 0.0002	0.4607 $\pm$ 0.0056	5
Reflow-1 (Euler)	Checkerboard	0.0018 $\pm$ 0.0004	0.0016 $\pm$ 0.0003	0.0017 $\pm$ 0.0003	0.0015 $\pm$ 0.0002	0.9988 $\pm$ 0.0004	5
Real data (floor)	Checkerboard	0.0005 $\pm$ 0.0001	0.0005 $\pm$ 0.0001	0.0005 $\pm$ 0.0001	0.0005 $\pm$ 0.0001	–	5

TABLE III

Paired tests over the five seeds, from rh compare: difference of the seed means (first system minus second) with the two-sided paired p in parentheses; for H2, the ratio of seed-mean SW at $K=4$ and at $K=100$. Anc.: DDPM ancestral.

Quantity	8 Gauss.	Moons	Checker.
Registered tests (5 seeds)			
H1 SW@4 DDIM-FM	0.202 (0.0005)	−0.060 (0.002)	0.013 (0.007)
H2 SW DDIM/FM, $K=4 \rightarrow 100$	3.34 $\rightarrow$ 2.15	0.77 $\rightarrow$ 1.25	1.06 $\rightarrow$ 1.73
H3 SW@1 FM-Reflow	1.192 (9.7×10^{-10})	1.117 (3.7×10^{-10})	0.899 (7.3×10^{-10})
H3 SW@100 FM-Reflow	−0.0010 (0.020)	−0.0012 (0.008)	−0.0003 (0.0002)
H4 straight FM-DDIM	−0.096 (6.7×10^{-5})	−0.138 (1.2×10^{-6})	−0.327 (8.1×10^{-7})
H4 straight FM-Reflow	−0.343 (1.5×10^{-10})	−0.323 (4.1×10^{-9})	−0.549 (2.8×10^{-9})
Exploratory, not registered (5 seeds)			
SW@100 FM-DDIM	−0.030 (0.086)	−0.006 (0.048)	−0.022 (9.7×10^{-5})
MMD@4 DDIM-FM	0.0341 (0.001)	0.0107 (8.2×10^{-5})	0.0066 (0.0003)
MMD@1 FM-DDIM	0.490 (7.5×10^{-6})	0.381 (3.6×10^{-6})	0.405 (1.4×10^{-6})
SW@4 DDIM-Anc.	−0.041 (0.068)	−0.252 (3.8×10^{-6})	−0.122 (1.5×10^{-5})
MMD@4 DDIM-Anc.	−0.0087 (0.050)	−0.0235 (5.3×10^{-6})	−0.0178 (1.5×10^{-5})

the DDIM seed spread is large (0.106 ± 0.052) and we do not claim a reliable difference there. The advantage disappears at low budgets: at 4 evaluations Heun is worse than Euler on eight Gaussians (0.138 against 0.086), and at 2 evaluations it is worse on all three tasks. From 8 evaluations on, a second-order solver on the unchanged flow model thus closes most of the gap to Reflow-1 (0.041 against 0.031 for Reflow-1 at $K=8$ on two moons) without reflow’s extra training. We ran no second-order solver for the diffusion model, so this is not a statement about flows against diffusion.

VI. Ablations and Analysis

against 0.054); its mean SW is also below that of DDIM at 8 evaluations on all three tasks, but on eight Gaussians

Three ablations (three seeds each, seeds 0–2) probe the reflow and the diffusion baseline; they were chosen before

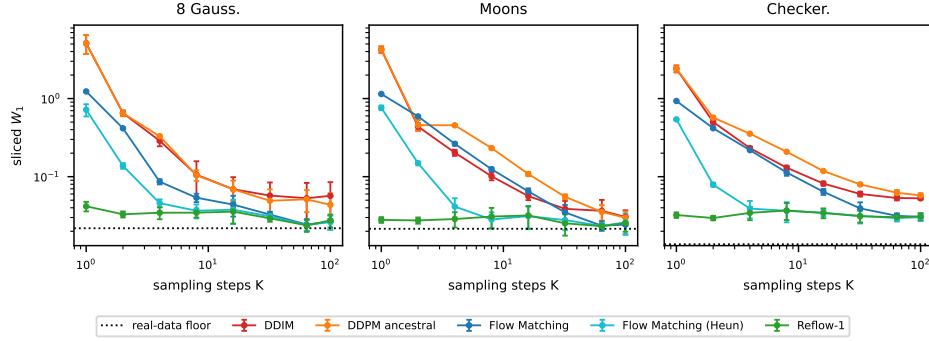


Fig. 1. Sliced W_1 against the number of sampling steps K (mean and std over five seeds; log axes). The dotted line is the real-data floor. Heun costs $2K$ evaluations.

running, and a fourth variant, the cosine-start diagnostic, was added after seeing the first cosine result.

Diffusion schedule. Replacing the linear β schedule by the cosine schedule of [10] makes DDIM fail without any $\hat{x}_0$ clipping (Table IV, top): even at $K=100$ the SW is above 1 on every task, and at $K=1$ it is several hundred. A diagnostic variant changes only the start index of the uniform grid (it starts at the last index with $\bar{\alpha} \geq 4 \times 10^{-5}$, about the linear schedule’s terminal value; same trained network): with that start index the 100-step SW drops to 0.054, 0.048 and 0.074. The failure is therefore tied to the first sampling step; it may arise because at $\bar{\alpha}_T \approx 0$ the estimate $\hat{x}_0$ divides the noise-prediction error by $\sqrt{\bar{\alpha}}$, which we did not test separately. The truncated-start variant is nominally better than linear-schedule DDIM at $K=1$ (within one standard deviation on the checkerboard, and we did not test matched seeds) but worse at $K=16$ on all three tasks and at $K=4$ on eight Gaussians and two moons (Tables I and IV, Fig. 2; seeds 0–2 against 0–4 for the linear row), so, with three seeds, the weak few-step DDIM baseline in Table I does not appear to be an artefact of choosing the linear schedule, although we did not tune other samplers or schedules.

Reflow teacher and second round. Generating the reflow pairs with a 4-step instead of a 100-step teacher caps the student near the teacher’s quality at every step count shown (Table IV, bottom: SW roughly constant in K , and close to FM-Euler’s $K=4$ SW in Table I); the straightness is still 0.996, 0.995 and 0.997, so a straight student can be a bad generator when its pairs come from a poor sampler. Set against Reflow-1 (Table I), the student of the 4-step teacher has the higher SW on every task at $K=1$ and at $K=100$ (on two moons at $K=1$, 0.269 against 0.028), while a second reflow round leaves SW at $K=1$ and $K=100$ close to that of Reflow-1 (on eight Gaussians at $K=1$, 0.033 against 0.042; at $K=100$, 0.029 against 0.028), for one more round of pair generation and training. These are comparisons of table means (seeds 0–2 for the ablations, 0–4 for Reflow-1); no paired test is reported for them.

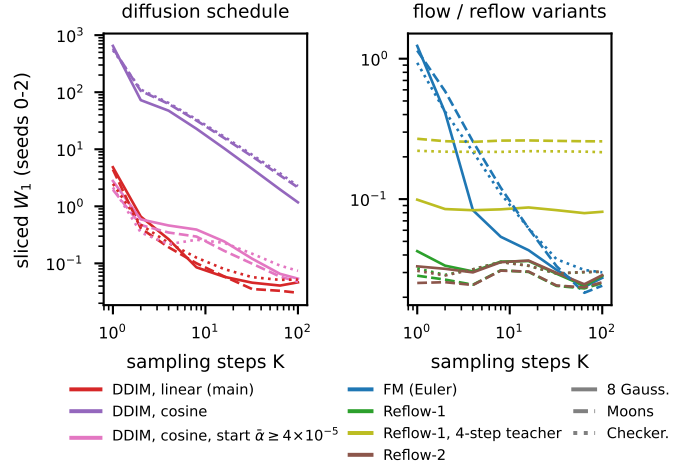


Fig. 2. Ablations (sliced W_1 , mean over seeds 0–2; line style = task). Left: diffusion schedule. Right: flow and reflow variants.

VII. Limitations

Scale. Three 2D densities, one MLP (four layers of 128), 3000 updates, five seeds (three for ablations): the paired tests have four degrees of freedom and are uncorrected for multiplicity, so p values are indicative, and the ablations are compared by their means only. Nothing here transfers to images or to larger networks without further evidence. **No tuning.** Every system used the same hyper-parameters, set once; a tuned diffusion baseline (other schedules, noise-level grids, higher-order solvers, $\hat{x}_0$ clipping, EDM-style preconditioning [11]) could close part of the gap, and the few-step diffusion deficit may be specific to the uniform timestep grid used. **Compute.** Reflow gets twice the updates of FM and 20000 pairs; we did not run an FM model trained for the same total. **Metrics.** SW with 256 fixed directions and a four-bandwidth biased MMD on 3000 points are noisy near the floor (the real-data floor is itself 0.014 to 0.022 SW), so differences among the best systems at $K \geq 16$ are not resolved; the two metrics rank FM and DDIM differently at $K=1$ on all tasks (SW favours FM, MMD favours DDIM) and at $K=4$ and 16

TABLE IV

Ablations (mean $\pm$ std over seeds 0–2). Top: cosine-schedule DDIM with and without the truncated start. Bottom: reflow with a 4-step teacher and a second reflow round (compare with Reflow-1 in Table I).

Method	Task	SW@1 $\downarrow$	SW@4 $\downarrow$	SW@16 $\downarrow$	SW@100 $\downarrow$	MMD@100 $\downarrow$	n
DDIM (cosine)	8 Gaussians	648.562 $\pm$ 105.784	47.441 $\pm$ 7.932	10.331 $\pm$ 1.775	1.174 $\pm$ 0.268	0.102 $\pm$ 0.033	3
DDIM (cosine, start at $\bar{\alpha} \geq 4 \times 10^{-5}$)	8 Gaussians	2.780 $\pm$ 0.588	0.461 $\pm$ 0.189	0.242 $\pm$ 0.067	0.054 $\pm$ 0.004	0.002 $\pm$ 0.000	3
DDIM (cosine)	Two moons	586.667 $\pm$ 58.955	62.387 $\pm$ 5.553	15.683 $\pm$ 2.192	2.142 $\pm$ 0.370	0.144 $\pm$ 0.021	3
DDIM (cosine, start at $\bar{\alpha} \geq 4 \times 10^{-5}$)	Two moons	1.952 $\pm$ 0.477	0.347 $\pm$ 0.072	0.166 $\pm$ 0.050	0.048 $\pm$ 0.015	0.002 $\pm$ 0.001	3
DDIM (cosine)	Checkerboard	545.477 $\pm$ 117.538	64.909 $\pm$ 16.613	16.593 $\pm$ 4.772	2.268 $\pm$ 0.599	0.184 $\pm$ 0.033	3
DDIM (cosine, start at $\bar{\alpha} \geq 4 \times 10^{-5}$)	Checkerboard	2.145 $\pm$ 0.786	0.217 $\pm$ 0.037	0.229 $\pm$ 0.078	0.074 $\pm$ 0.017	0.005 $\pm$ 0.002	3

Method	Task	SW@1 $\downarrow$	SW@4 $\downarrow$	SW@16 $\downarrow$	SW@100 $\downarrow$	straight $\uparrow$	n
Reflow-1 (teacher 4 steps)	8 Gaussians	0.099 $\pm$ 0.004	0.083 $\pm$ 0.004	0.087 $\pm$ 0.007	0.081 $\pm$ 0.002	0.996 $\pm$ 0.001	3
Reflow-2 (Euler)	8 Gaussians	0.033 $\pm$ 0.007	0.030 $\pm$ 0.002	0.037 $\pm$ 0.009	0.029 $\pm$ 0.006	1.000 $\pm$ 0.000	3
Reflow-1 (teacher 4 steps)	Two moons	0.269 $\pm$ 0.001	0.256 $\pm$ 0.005	0.262 $\pm$ 0.005	0.258 $\pm$ 0.008	0.995 $\pm$ 0.001	3
Reflow-2 (Euler)	Two moons	0.025 $\pm$ 0.003	0.024 $\pm$ 0.004	0.030 $\pm$ 0.008	0.026 $\pm$ 0.008	1.000 $\pm$ 0.000	3
Reflow-1 (teacher 4 steps)	Checkerboard	0.221 $\pm$ 0.006	0.217 $\pm$ 0.006	0.220 $\pm$ 0.007	0.216 $\pm$ 0.006	0.997 $\pm$ 0.001	3
Reflow-2 (Euler)	Checkerboard	0.031 $\pm$ 0.002	0.032 $\pm$ 0.001	0.034 $\pm$ 0.007	0.030 $\pm$ 0.001	1.000 $\pm$ 0.000	3

on two moons (SW favours DDIM, MMD favours FM, nominally at 16). Reimplementation. All methods are our own reimplementations, not the authors’ code. Process. A double launch of the run driver produced 88 duplicate registry rows (87 identical, one failed with empty metrics after a race on shared file paths); they were moved from results/runs.jsonl to a separate file, and four kept rows carry a timing measured in the sibling process. No run was dropped for its result. Mechanistic readings (the cosine failure, the teacher cap, the $K=1$ metric disagreement) rest on one diagnostic configuration each or on the table values only, not on dedicated interventions beyond those reported. Solvers. Heun was run only for the flow model; no second-order or multistep solver was run for the diffusion model, the Heun comparison and the exploratory tests of Table III were not registered in advance, and the Heun and reflow-ablation comparisons are reported as differences of seed means without a paired test.

VIII. Conclusion

With a shared MLP and no tuning, flow matching has lower SW than DDIM at one sampling step on three 2D densities but higher one-step MMD; at four steps it has lower SW on eight Gaussians, marginally lower SW on the checkerboard, and higher SW on two moons, while MMD ranks it ahead on all three tasks; the change of its advantage with the step count is mixed (the DDIM/FM SW ratio shrinks on eight Gaussians, grows on the checkerboard and changes from DDIM-better to FM-better on two moons), and by chord/arc length its paths are less straight than DDIM paths. One reflow round gives near-floor quality at one step, no gain at 100 steps, and a quality ceiling set by the teacher’s own sampler, with a second round adding little; without any retraining, Heun on the flow-matching model has lower SW than Euler at 8 network evaluations on all three tasks (by a small margin on eight Gaussians) and higher SW at 2. These are small-scale, five-seed observations; a full study would add tuned diffusion samplers, equal-compute reflow controls, more seeds and image-scale models.

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